Uncertainty Analysis Using PEcAn

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1. Introduction

Welcome to this PEcAn workflow notebook! This notebook will guide you through running an ecosystem model using PEcAn's programmatic interface.

1.1 What Is PEcAn?

PEcAn (Predictive Ecosystem Analyzer) is a scientific workflow system designed to make ecosystem modeling more transparent, repeatable, and accessible. It helps researchers:

- Run ecosystem models with standardized inputs and outputs
- Perform uncertainty analysis on model parameters
- Compare model predictions with observations
- Share and reproduce scientific workflows

1.2 What This Notebook Does

This notebook demonstrates how to:

1. Set up and configure a PEcAn workflow
2. Run an ecosystem model simulation
3. Perform ensemble-based analysis to quantify uncertainty in model outputs
4. Conduct sensitivity analysis to identify which parameters most influence model results
5. Analyze and visualize the results

1.3 Ensemble and Sensitivity Analysis in PEcAn

PEcAn can use information about parameter uncertainty to perform automated analyses:

- **Ensemble Analysis**: Repeats the model run many times, each with parameters sampled from their uncertainty distributions, to generate a probability distribution of model projections. This allows you to quantify confidence intervals on model outputs.
- **Sensitivity Analysis**: Systematically varies model parameters to assess how changes in each parameter affect model outputs. This identifies which parameters the model is most sensitive to.
- **Uncertainty Analysis**: Combines sensitivity and parameter uncertainty to determine the contribution of each parameter to the overall uncertainty in model outputs, helping to target parameters for future constraint.

These analyses are essential for understanding model behavior, quantifying uncertainty, and prioritizing data collection or parameter refinement.

1.4 Prerequisites

Before running this notebook, make sure you have:

- All the PEcAn packages installed. You can install all PEcAn packages and their dependencies by running the following command in the root of your PEcAn repository:

```
# Enable repository from pecanproject
options(repos = c(
  pecanproject = 'https://pecanproject.r-universe.dev',
  CRAN = 'https://cloud.r-project.org'))
# Download and install PEcAn.all in R
install.packages('PEcAn.all')
```

Before running this notebook, make sure you have:

- A valid `pecan.xml` configuration file or use the example provided: `pecan/documentation/tutorials/Demo`

1.5 How to Use This Notebook

1. Each section is clearly marked with a heading
2. Code chunks are provided with explanations
3. You can run the code chunks sequentially
4. Once you have successfully run the demo, you can modify parameters to configure new runs and analyses

Objective:

This demo illustrates how to run a basic PEcAn workflow using an R-based Quarto notebook. It will cover loading settings, writing model configuration files, and running model simulations. This approach provides a programmatic alternative to the web-based PEcAn interface for executing ecosystem models.

2. PEcAn Version

This sectioon prints PEcAn version for all the PEcAn packages.

```r
PEcAn.all::pecan_version()
```

```
 package              v1.9.0 installed source
 PEcAn.all            1.9.0  1.9.0     https://pecanproj...
 PEcAn.allometry      1.7.4  NA        NA
 PEcAn.assim.batch    1.9.0  1.9.0     https://pecanproj...
 PEcAn.BASGRA         1.8.1  NA        NA
 PEcAn.benchmark      1.7.4  1.7.4     local
 PEcAn.BIOCRO         1.7.4  1.7.4     https://pecanproj...
 PEcAn.CABLE          1.7.4  NA        NA
 PEcAn.CLM45          1.7.4  NA        NA
 PEcAn.DALEC          1.7.4  NA        NA
 PEcAn.data.atmosphere 1.9.0  1.9.0     local
 PEcAn.data.land      1.8.1  1.8.1     local
 PEcAn.data.mining    1.7.4  NA        NA
 PEcAn.data.remote    1.9.0  1.9.0     local
 PEcAn.DB             1.8.1  1.8.1     local (/home/arit...
 PEcAn.dvmdostem      1.7.4  NA        NA
 PEcAn.ED2            1.8.1  NA        NA
 PEcAn.emulator       1.8.1  1.8.1     local
```

```
 PEcAn.FATES            1.8.0  NA         NA
 PEcAn.GDAY             1.7.4  NA         NA
 PEcAn.JULES            1.7.4  NA         NA
 PEcAn.LDNDC            1.0.1  NA         NA
 PEcAn.LINKAGES         1.7.4  NA         NA
 PEcAn.logger           1.8.3  1.8.3      local (/home/arit...
 PEcAn.LPJGUESS         1.8.0  NA         NA
 PEcAn.MA               1.7.4  1.7.4      local
 PEcAn.MAAT             1.7.4  NA         NA
 PEcAn.MAESPA           1.7.4  NA         NA
 PEcAn.ModelName        1.8.1  1.8.1      local (/home/arit...
 PEcAn.photosynthesis   1.7.4  NA         NA
 PEcAn.PRELES           1.7.4  NA         NA
 PEcAn.priors           1.7.4  1.7.4      local
 PEcAn.qaqc             1.7.4  NA         NA
 PEcAn.remote           1.9.0  1.9.0      local (/home/arit...
 PEcAn.settings         1.9.0  1.9.1      local (/home/arit...
 PEcAn.SIBCASA          0.0.2  NA         NA
 PEcAn.SIPNET           1.9.0  1.9.1      local
 PEcAn.STICS            1.8.1  NA         NA
 PEcAn.uncertainty      1.8.1  1.8.1      local
 PEcAn.utils            1.8.1  1.8.1      local (/home/arit...
 PEcAn.visualization    1.8.1  1.8.1      local (/home/arit...
 PEcAn.workflow         1.9.0  1.9.0      local
 PEcAnAssimSequential   1.9.0  NA         NA
 PEcAnRTM               1.7.4  NA         NA
```

3. Install SIPNET v1.3.0

If you haven't already installed the SIPNET binary, you can do so by running the following code. This will download the SIPNET binary to `demo_outdir/sipnet` and make it executable.

> Note: The `demo_outdir` directory will be created in the root of your PEcAn installation (i.e., at `pecan/demo_outdir/`). This directory will contain the SIPNET binary as well as the output generated by PEcAn in this demo.

```r
# Download and install SIPNET v1.3.0
source(
  here::here(
    "documentation/tutorials/Demo_1_Basic_Run/download_sipnet.R"
  )
)
```

Note: You can find the most recent version of the SIPNET binary at: SIPNET GitHub Releases, but this notebook is designed to work with SIPNET v1.3.0.

2. Load PEcAn Packages

First, we need to load the PEcAn R packages. These packages provide all the functions we'll use to run the workflow.

```r
# Load the PEcAn.all package, which includes all necessary PEcAn functionality
library("PEcAn.all")
```

```
Loading required package: PEcAn.DB

Loading required package: PEcAn.settings

Loading required package: PEcAn.MA

Loading required package: PEcAn.logger

Loading required package: PEcAn.utils

Loading required package: PEcAn.uncertainty

Loading required package: PEcAn.data.atmosphere

Loading required package: PEcAn.data.land

Loading required package: PEcAn.data.remote

Loading required package: PEcAn.assim.batch

Loading required package: PEcAn.emulator

Loading required package: PEcAn.priors

Loading required package: PEcAn.benchmark

Loading required package: PEcAn.remote

Loading required package: PEcAn.workflow
```

3. Load PEcAn Settings Files

PEcAn uses an XML-based settings file (pecan.xml) to configure model runs. This file defines key information about the run including: PFT(s), site location, time period of the run, the location of input files and where outputs will be saved. Other settings outside the scope of this demo include the types of analyses that will be performed, how to connect to a database, and how to run it on a high performance computing cluster (we are using the default single model run on a single computer).

You can read more about the settings file in the "PEcAn XML" chapter of the documentation.

There is an example `pecan.xml` that has been configured for this demonstration. You can find it at `pecan/documentation/tutorials/Demo_2_Uncertainty_Analysis/pecan.xml`.

This is how PEcAn loads the settings file:

```
settings_path <- here::here("documentation/tutorials/Demo_02_Uncertainty_Analysis/pecan.xml")
```

4. Prepare and Validate Settings

After specifying the path to your `pecan.xml` file, the next step involves reading and preparing these settings. PEcAn provides utilities to process and validate your configurations before any execution begins.

- `PEcAn.settings::read.settings(settings_path)`: Parses the `pecan.xml` file, converting its contents into an R list object that PEcAn can work with.
- `PEcAn.settings::prepare.settings(settings)`: Further prepares and validates the settings, checking for missing required fields and ensuring consistency.

```
# Read the settings from the pecan.xml file
settings <- PEcAn.settings::read.settings(settings_path)

# Prepare and validate the settings
settings <- PEcAn.settings::prepare.settings(settings)
```

5. Explore the Settings Object

Once the settings have been read and prepared, it is useful to inspect the structure of the `settings` object. This object is an R list containing all parameters and configurations for the PEcAn workflow.

- `str(settings)` displays the internal structure of the `settings` object. This shows how the settings are represented in R and is useful for debugging and verifying settings.

```r
str(settings)
```

```
List of 13
 $ info                :List of 4
  ..$ notes   : NULL
  ..$ userid  : chr "-1"
  ..$ username: NULL
  ..$ date    : chr "2025-06-19-15-34-01"
 $ outdir              : chr "/home/aritra/Downloads/projects/gsoc-codebases/pecan/documenta
 $ pfts                :List of 1
  ..$ pft:List of 3
  .. ..$ name           : chr "temperate.coniferous"
  .. ..$ posterior.files: chr "pft/temperate.coniferous/prior.distns.Rdata"
  .. ..$ outdir         : chr "pft/temperate.coniferous/pft/temperate.coniferous"
 $ ensemble            :List of 5
  ..$ size         : chr "10"
  ..$ variable     : chr "NPP"
  ..$ samplingspace:List of 1
  .. ..$ parameters:List of 1
  .. .. ..$ method: chr "uniform"
  ..$ start.year   : chr "2004"
  ..$ end.year     : chr "2004"
 $ sensitivity.analysis:List of 4
  ..$ quantiles :List of 2
  .. ..$ sigma: chr "-1"
  .. ..$ sigma: chr "1"
  ..$ variable  : chr "NPP"
  ..$ start.year: chr "2004"
  ..$ end.year  : chr "2004"
 $ model               :List of 5
  ..$ type      : chr "SIPNET"
  ..$ revision  : chr "git"
  ..$ delete.raw: chr "FALSE"
  ..$ binary    : chr "demo_outdir/sipnet"
  ..$ id        : num -1
 $ run                 :List of 4
  ..$ site      :List of 6
  .. ..$ met.start: chr "2004/01/01"
  .. ..$ met.end  : chr "2004/12/31"
```

```
  .. ..$ name     : chr "Niwot Ridge Forest/LTER NWT1 (US-NR1)"
  .. ..$ lat      : chr "40.0329"
  .. ..$ lon      : chr "-105.546"
  .. ..$ id       : num -1
  ..$ inputs    :List of 1
  .. ..$ met:List of 4
  .. .. ..$ source  : chr "AmerifluxLBL"
  .. .. ..$ output  : chr "SIPNET"
  .. .. ..$ username: chr "Aritra_2004"
  .. .. ..$ path    :List of 1
  .. .. .. ..$ path1: chr "dbfiles/AMF_US-NR1_BASE_HH_23-5.2004-01-01.2004-12-31.clim"
  ..$ start.date: chr "2004/01/01"
  ..$ end.date  : chr "2004/12/31"
 $ host                :List of 3
  ..$ name  : chr "localhost"
  ..$ rundir: chr "/home/aritra/Downloads/projects/gsoc-codebases/pecan/documentation/tutoria
  ..$ outdir: chr "/home/aritra/Downloads/projects/gsoc-codebases/pecan/documentation/tutoria
 $ settings.info       :List of 3
  ..$ deprecated.settings.fixed: logi TRUE
  ..$ settings.updated         : logi TRUE
  ..$ checked                  : logi TRUE
 $ database            :List of 1
  ..$ dbfiles: chr "/home/aritra/.pecan/dbfiles"
 $ workflow            :List of 1
  ..$ id: chr "2025-08-27-01-29-43"
 $ rundir              : chr "/home/aritra/Downloads/projects/gsoc-codebases/pecan/documenta
 $ modeloutdir         : chr "/home/aritra/Downloads/projects/gsoc-codebases/pecan/documenta
 - attr(*, "class")= chr [1:3] "Settings" "SafeList" "list"
```

Your turn: explore the `settings` object further using the following commands: * `names(settings)` to list the top-level keys in the settings object * Once you know the names, you can look at each component in detail, for example: * `settings$run` to access the run-specific settings, such as start and end dates, model type, and output directory * `settings$pfts` to explore the Plant Functional Types settings

Now you can update each of these settings. Here is a simple example:

```r
settings$info <- list(
  author = "Aritra Dey",
  date = Sys.Date(),
  description = "Demo run of PEcAn using SIPNET"
)
```

Editing the more interesting settings to change the PFT (`settings$pfts`) or extend the run (`settings$run$end.date`) is beyond the scope of this demo. You *could* change the pft or the end date, but you would need a new file containing parameters for that PFT (`settings$pfts$pft$posterior.files`), or a climate file (`settings$run$met$path$path1`) that extends to the desired simulation period.

8. Create Output Directory

Before running the workflow, we need to ensure that the output directory specified in the settings exists. This directory will store all the model outputs, configuration files, and results from the simulation.

> **Note:** All outputs will be written to `settings$outdir`. You can change this directory in your `pecan.xml` file or by modifying `settings$outdir` in the notebook before running the workflow. You can also put the `sipnet` executable anywhere you prefer, as long as you update `settings$model$binary`.

```r
dir.create(settings$outdir, recursive = TRUE, showWarnings = FALSE)
```

9. Write Model Configuration Files

This step generates the model-specific configuration files that will be used to run the ecosystem model. The process involves:

1. Disabling database write operations because we are not using a database
2. Generating SIPNET configuration files using the `runModule.run.write.configs()` function.

```r
settings$database <- NULL # Disable database operations for this demo
settings <- PEcAn.workflow::runModule.run.write.configs(settings)
```

```
Error in postgresqlNewConnection(drv, ...) :
  RPosgreSQL error: could not connect aritra@/var/run/postgresql:5432 on dbname "aritra": co
    Is the server running locally and accepting connections on that socket?
```

```
Loading required package: PEcAn.SIPNET
```

```
Warning in dir.create(file.path(settings$rundir, run.id), recursive = TRUE):
'/home/aritra/Downloads/projects/gsoc-codebases/pecan/documentation/tutorials/Demo_02_Uncerta
already exists

Warning in dir.create(file.path(settings$modeloutdir, run.id), recursive =
TRUE):
'/home/aritra/Downloads/projects/gsoc-codebases/pecan/documentation/tutorials/Demo_02_Uncerta
already exists

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already exists
```

10. Run Model Simulations and Fetch Results

This section executes the actual model simulations and retrieves the results. The process is managed by PEcAn's workflow system, which handles the execution of your chosen ecosystem model.

- `runModule_start_model_runs(settings)`: This function initiates the model runs based on your configuration. It manages the execution of your chosen ecosystem model, using the configuration files generated in the previous step.

```
PEcAn.workflow::runModule_start_model_runs(settings)
```

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```

10. Ensemble and Sensitivity Analysis

This section executes ensemble and sensitivity analyses to comprehensively assess model uncertainty and parameter importance.

Ensemble Analysis: Quantifies uncertainty in model predictions by running multiple simulations with parameters sampled from their uncertainty distributions. This generates probability distributions of model outputs and confidence intervals.

Sensitivity Analysis: Systematically varies individual parameters to assess their influence on model outputs. This identifies which parameters most strongly affect model predictions and helps prioritize parameter refinement efforts.

```r
# Execute ensemble and sensitivity analyses to quantify uncertainty and parameter influence
# First, retrieve results from all model runs
runModule.get.results(settings)

# Run ensemble analysis: generates ensemble predictions and uncertainty quantification
runModule.run.ensemble.analysis(settings)
```

```
[1] "----- Variable: NPP"
[1] "----- Running ensemble analysis for site:  Niwot Ridge Forest/LTER NWT1 (US-NR1)"

[1] "----- Done!"
[1] " "
[1] "----------------------------------------------"
[1] " "
[1] " "
```

```r
# Run sensitivity analysis: generates variance decomposition plots and sensitivity metrics
runModule.run.sensitivity.analysis(settings)
```

```
$coef.vars
        growth_resp_factor         leaf_turnover_rate
                0.52300833                 0.86632090
     root_respiration_rate         root_turnover_rate
                0.57788540                 0.57437841
                      Amax     leaf_respiration_rate_m2
                0.57908491                 0.54901075
                       SLA                      leafC
                0.82117951                 0.02619998
               Vm_low_temp                   AmaxFrac
                0.01100267                 0.11526796
                   psnTOpt      stem_respiration_rate
                0.03421984                 0.57039501
   extinction_coefficient         half_saturation_PAR
                0.13867237                 0.43358392
                 dVPDSlope                    dVpdExp
                0.53245546                 0.28851258
       veg_respiration_Q10    fine_root_respiration_Q10
                0.17166374                 0.32637873
coarse_root_respiration_Q10
                0.32456556
```

```
$elasticities
        growth_resp_factor           leaf_turnover_rate
                0.00000000                   0.03160548
      root_respiration_rate           root_turnover_rate
               -0.02584500                   0.05423150
                      Amax     leaf_respiration_rate_m2
                2.86845928                  -2.18855458
                       SLA                        leafC
                0.78578615                   0.22412360
               Vm_low_temp                     AmaxFrac
                1.01784423                   2.54076415
                   psnTOpt        stem_respiration_rate
                1.01458549                  -1.49110518
    extinction_coefficient            half_saturation_PAR
               -0.58959109                  -1.29916104
                 dVPDSlope                      dVpdExp
               -0.18711936                   0.09862586
        veg_respiration_Q10    fine_root_respiration_Q10
                6.65265264                   0.08805556
coarse_root_respiration_Q10
               -0.24384160

$sensitivities
        growth_resp_factor           leaf_turnover_rate
              0.000000e+00                 1.311650e-10
      root_respiration_rate           root_turnover_rate
             -2.351787e-12                 4.974212e-11
                      Amax     leaf_respiration_rate_m2
              6.620714e-10                -1.991476e-09
                       SLA                        leafC
              2.508045e-10                 2.035098e-11
               Vm_low_temp                     AmaxFrac
             -9.552636e-10                 1.554780e-08
                   psnTOpt        stem_respiration_rate
              2.506784e-10                -1.355327e-10
    extinction_coefficient            half_saturation_PAR
             -5.411301e-09                -3.888258e-10
                 dVPDSlope                      dVpdExp
             -6.606424e-09                 2.252074e-10
        veg_respiration_Q10    fine_root_respiration_Q10
              1.526329e-08                 1.268657e-10
coarse_root_respiration_Q10
             -3.502367e-10
```

```
$variances
         growth_resp_factor          leaf_turnover_rate
               1.847507e-50                2.362458e-20
      root_respiration_rate          root_turnover_rate
               4.733959e-21                2.250602e-20
                       Amax    leaf_respiration_rate_m2
               5.978007e-17                2.921097e-17
                        SLA                       leafC
               1.287472e-17                7.771514e-22
                Vm_low_temp                    AmaxFrac
               8.240993e-18                1.813578e-18
                    psnTOpt       stem_respiration_rate
               6.435354e-18                1.525807e-17
     extinction_coefficient          half_saturation_PAR
               1.410554e-19                6.701214e-18
                  dVPDSlope                     dVpdExp
               2.115325e-19                1.757074e-20
        veg_respiration_Q10   fine_root_respiration_Q10
               2.807372e-17                1.901468e-20
coarse_root_respiration_Q10
               1.321859e-19

$partial.variances
         growth_resp_factor          leaf_turnover_rate
               1.093448e-34                1.398221e-04
      root_respiration_rate          root_turnover_rate
               2.801794e-05                1.332019e-04
                       Amax    leaf_respiration_rate_m2
               3.538084e-01                1.728851e-01
                        SLA                       leafC
               7.619906e-02                4.599572e-06
                Vm_low_temp                    AmaxFrac
               4.877433e-02                1.073366e-02
                    psnTOpt       stem_respiration_rate
               3.808765e-02                9.030490e-02
     extinction_coefficient          half_saturation_PAR
               8.348364e-04                3.966114e-02
                  dVPDSlope                     dVpdExp
               1.251955e-03                1.039925e-04
        veg_respiration_Q10   fine_root_respiration_Q10
               1.661544e-01                1.125384e-04
coarse_root_respiration_Q10
```

```
           7.823427e-04

       growth_resp_factor leaf_turnover_rate root_respiration_rate
15.866       4.593895e-09       4.311343e-09          4.684939e-09
50           4.593895e-09       4.593895e-09          4.593895e-09
84.134       4.593895e-09       4.644113e-09          4.521637e-09
       root_turnover_rate          Amax leaf_respiration_rate_m2          SLA
15.866       4.328356e-09 -7.306118e-09             9.968630e-09 7.099398e-10
50           4.593895e-09  4.593895e-09             4.593895e-09 4.593895e-09
84.134       4.664352e-09  1.065362e-08            -1.113072e-09 8.332701e-11
              leafC  Vm_low_temp     AmaxFrac       psnTOpt
15.866 4.556359e-09 6.867663e-09 2.897566e-09 -5.646314e-09
50     4.593895e-09 4.593895e-09 4.593895e-09  4.593895e-09
84.134 4.610055e-09 1.709548e-09 6.063803e-09  7.579564e-09
       stem_respiration_rate extinction_coefficient half_saturation_PAR
15.866          9.194242e-09           5.020966e-09        8.024502e-09
50              4.593895e-09           4.593895e-09        4.593895e-09
84.134         -3.140179e-11           4.133589e-09        1.921929e-09
          dVPDSlope      dVpdExp veg_respiration_Q10 fine_root_respiration_Q10
15.866 5.035481e-09 4.396093e-09       -3.096286e-09              4.349512e-09
50     4.593895e-09 4.593895e-09        4.593895e-09              4.593895e-09
84.134 3.953337e-09 4.706347e-09        9.318616e-09              4.661316e-09
       coarse_root_respiration_Q10
15.866                5.027164e-09
50                    4.593895e-09
84.134                4.167868e-09

TableGrob (5 x 4) "arrange": 19 grobs
                          z     cells    name           grob
growth_resp_factor        1 (1-1,1-1) arrange gtable[layout]
leaf_turnover_rate        2 (1-1,2-2) arrange gtable[layout]
root_respiration_rate     3 (1-1,3-3) arrange gtable[layout]
root_turnover_rate        4 (1-1,4-4) arrange gtable[layout]
Amax                      5 (2-2,1-1) arrange gtable[layout]
leaf_respiration_rate_m2  6 (2-2,2-2) arrange gtable[layout]
SLA                       7 (2-2,3-3) arrange gtable[layout]
leafC                     8 (2-2,4-4) arrange gtable[layout]
Vm_low_temp               9 (3-3,1-1) arrange gtable[layout]
AmaxFrac                 10 (3-3,2-2) arrange gtable[layout]
psnTOpt                  11 (3-3,3-3) arrange gtable[layout]
stem_respiration_rate    12 (3-3,4-4) arrange gtable[layout]
extinction_coefficient   13 (4-4,1-1) arrange gtable[layout]
```

```
half_saturation_PAR         14 (4-4,2-2) arrange gtable[layout]
dVPDSlope                   15 (4-4,3-3) arrange gtable[layout]
dVpdExp                     16 (4-4,4-4) arrange gtable[layout]
veg_respiration_Q10         17 (5-5,1-1) arrange gtable[layout]
fine_root_respiration_Q10   18 (5-5,2-2) arrange gtable[layout]
coarse_root_respiration_Q10 19 (5-5,3-3) arrange gtable[layout]
$growth_resp_factor


$leaf_turnover_rate


$root_respiration_rate


$root_turnover_rate


$Amax


$leaf_respiration_rate_m2


$SLA


$leafC


$Vm_low_temp


$AmaxFrac


$psnTOpt


$stem_respiration_rate
```

```
$extinction_coefficient
```

```
$half_saturation_PAR
```

```
$dVPDSlope
```

```
$dVpdExp
```

```
$veg_respiration_Q10
```

```
$fine_root_respiration_Q10
```

```
$coarse_root_respiration_Q10
```

11. Understanding PEcAn Uncertainty Analysis Outputs

After running ensemble and sensitivity analyses, PEcAn generates several important outputs that help you understand model uncertainty and parameter sensitivity.

11.1 Ensemble Analysis Outputs

The ensemble analysis produces:

- **`ensemble.Rdata`**: Contains `ensemble.output` object with model predictions for all ensemble members
- **`ensemble.analysis.[RunID].[Variable].[StartYear].[EndYear].pdf`**: Histogram and boxplot of ensemble predictions
- **`ensemble.ts.[RunID].[Variable].[StartYear].[EndYear].pdf`**: Time-series plot showing ensemble mean, median, and 95% confidence intervals

11.2 Sensitivity Analysis Outputs

The sensitivity analysis generates:

- **`sensitivity.analysis.[RunID].[Variable].[StartYear].[EndYear].pdf`**: Raw data points from univariate analyses with spline fits
- **`sensitivity.output.[RunID].[Variable].[StartYear].[EndYear].Rdata`**: Model outputs corresponding to parameter variations

11.3 Uncertainty Analysis Outputs

The uncertainty analysis produces:

- **`variance.decomposition.[RunID].[Variable].[StartYear].[EndYear].pdf`**: Three-column analysis showing:
  - Coefficient of variation (normalized posterior variance)
  - Elasticity (normalized sensitivity)
  - Partial standard deviation of each parameter

11.4 Interpreting the Results

Variance Decomposition Analysis: - Parameters are sorted by their contribution to model output variability - Identify top-tier parameters for future constraint - Distinguish between parameters that are: - Highly sensitive but low uncertainty - Highly uncertain but low sensitivity
- Both sensitive and uncertain (priority targets)

Parameter Prioritization: - Focus on parameters that are both sensitive and uncertain - Consider measurement costs and feasibility - Balance parameter uncertainty vs. model sensitivity

12. Customizing Ensemble Analysis Parameters (Optional)

(Optional) Use this section only if you want to override the default ensemble analysis parameters. Skip if defaults are sufficient.

> **Important:** If you modify the ensemble analysis parameters in this section, you must re-run sections 8, 9, and 10 to regenerate the model output with the new ensemble configuration.

```
# Set the number of ensemble members (model runs)
settings$ensemble$size <- 50

# Specify the variable to be analyzed in the ensemble
settings$ensemble$variable <- "NEE"

# Choose the method for sampling the parameter space (options: "uniform", "lhc", "halton", "s
settings$ensemble$samplingspace$parameters$method <- "halton"
```

13. Customizing Sensitivity Analysis Parameters (Optional)

(Optional) Use this section only if you want to override the default sensitivity analysis parameters. Skip if defaults are sufficient.

> **Important:** If you modify the sensitivity analysis parameters in this section, you must re-run sections 8, 9, and 10 to regenerate the model output with the new sensitivity analysis configuration.

```
# Set the quantiles (in standard deviations) for parameter distribution in sensitivity analys
settings$sensitivity.analysis$quantiles$sigma <- c(-2, -1, 1, 2)

# Specify the variable to be analyzed in sensitivity analysis
settings$sensitivity.analysis$variable <- "NEE"
```

14. Extract Model Results and Prepare for Analysis

After the model simulation completes, we need to extract the results and prepare them for analysis. This involves:

1. Reading the run ID
2. Setting up output paths
3. Defining time period
4. Loading model output

```
runid <- as.character(read.table(paste(settings$outdir, "/run/", "runs.txt", sep = ""))[1, 1]
# You can change [1,1] to [10,1], [5,1], etc. to select different run IDs from runs.txt
# For example: [10,1] selects the 10th run ID, [5,1] selects the 5th run ID
outdir <- paste(settings$outdir, "/out/", runid, sep = "")
start.year <- as.numeric(lubridate::year(settings$run$start.date))
```

```r
end.year <- as.numeric(lubridate::year(settings$run$end.date))
model_output <- PEcAn.utils::read.output(
  runid,
  outdir,
  start.year,
  end.year,
  variables = NULL,
  dataframe = TRUE,
  verbose = FALSE
)
available_vars <- names(model_output)[!names(model_output) %in% c("posix", "time_bounds")]
```

15. Display Available Model Variables

This section shows all the variables that are available in the model output. These variables represent different ecosystem processes and states that the model has simulated.

```r
vars_df <- PEcAn.utils::standard_vars |>
  dplyr::select(
    Variable = Variable.Name,
    Description = Long.name
  ) |>
  dplyr::filter(Variable %in% available_vars) |>
  # TODO: add year to PEcAn.utils::standard vars
  dplyr::bind_rows(
    dplyr::tibble(
      Variable = "year",
      Description = "Year"
    )
  )

vars_df$Description[is.na(vars_df$Description)] <- "(No description available)"
knitr::kable(vars_df, caption = "Model Output Variables and Descriptions")
```

Table 1: Model Output Variables and Descriptions

Variable	Description
GPP	Gross Primary Productivity
NEE	Net Ecosystem Exchange

Variable	Description
TotalResp	Total Respiration
AutoResp	Autotrophic Respiration
HeteroResp	Heterotrophic Respiration
SoilResp	Soil Respiration
NPP	Net Primary Productivity
GWBI	Gross Woody Biomass Increment
TotLivBiom	Total living biomass
AGB	Total aboveground biomass
LAI	Leaf Area Index
leaf_carbon_content	Leaf Carbon Content
fine_root_carbon_content	Fine Root Carbon Content
coarse_root_carbon_content	Coarse Root Carbon Content
AbvGrndWood	Above ground woody biomass
TotSoilCarb	Total Soil Carbon
litter_carbon_content	Litter Carbon Content
Qle	Latent heat
Transp	Total transpiration
SoilMoist	Average Layer Soil Moisture
SoilMoistFrac	Average Layer Fraction of Saturation
SWE	Snow Water Equivalent
litter_mass_content_of_water	Average layer litter moisture
year	Year

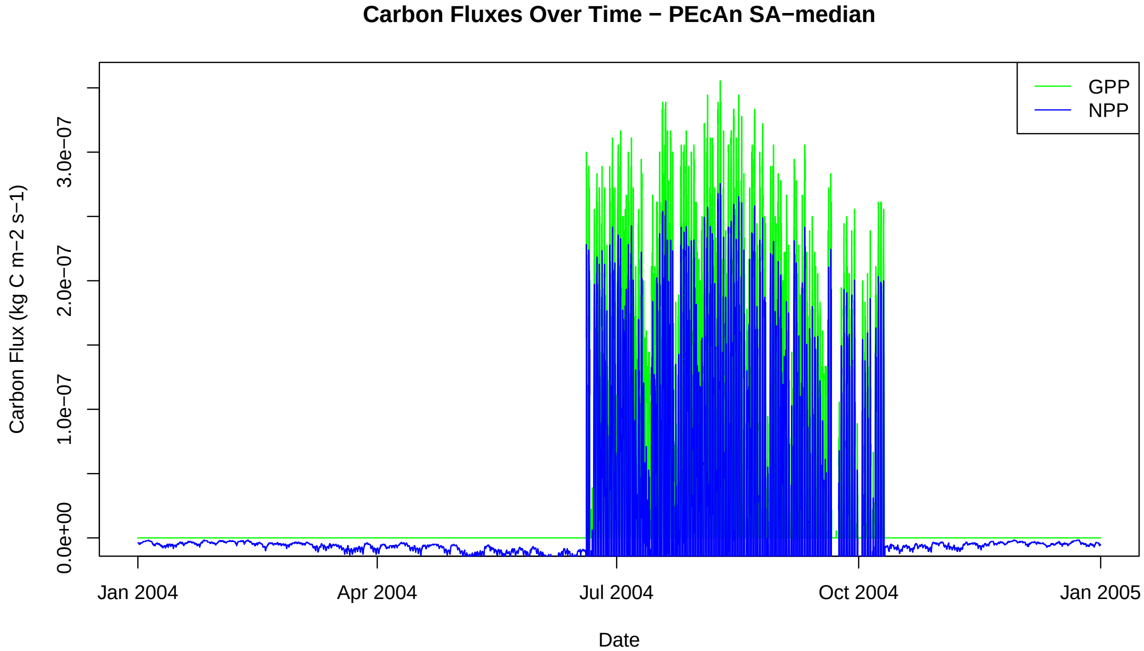
16. Visualize Model Results

This section provides examples of how to create time series plots of different model variables. The examples cover various ecosystem processes including carbon fluxes, carbon pools, water variables, and structural variables like Leaf Area Index (LAI).

16.1 Plot Carbon Fluxes

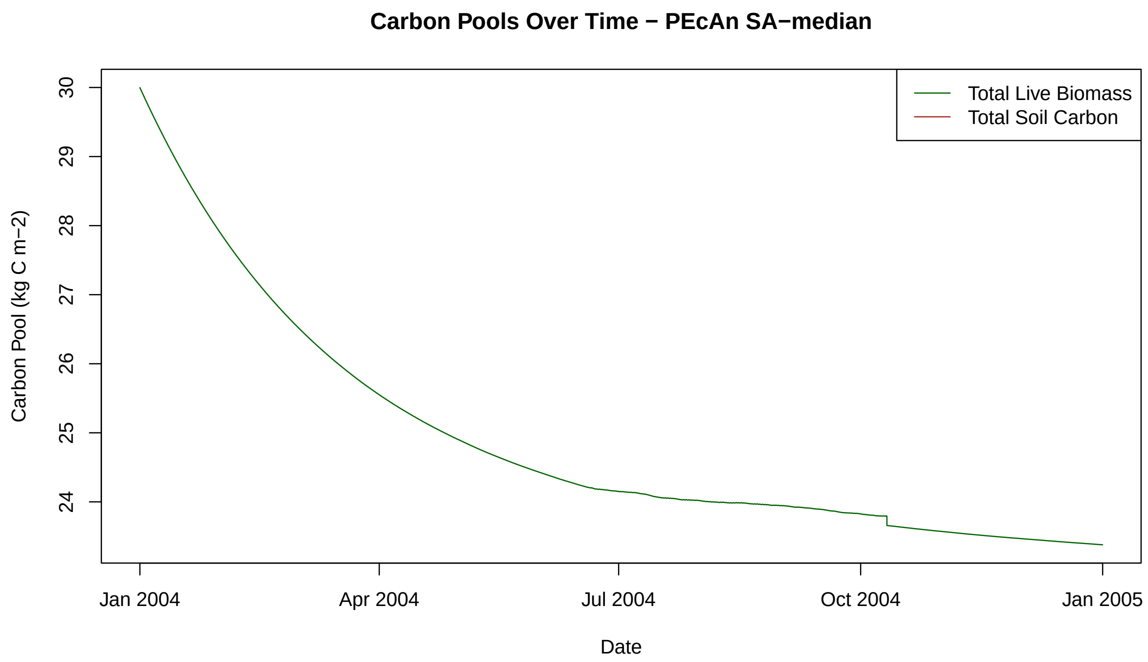
```
# Plot Gross Primary Productivity (GPP) and Net Primary Productivity (NPP)
plot(model_output$posix, model_output$GPP,
  type = "l",
  col = "green",
  xlab = "Date",
  ylab = "Carbon Flux (kg C m-2 s-1)",
```

```r
  main = paste("Carbon Fluxes Over Time - PEcAn", runid)
)
lines(model_output$posix, model_output$NPP, col = "blue")
legend("topright", legend = c("GPP", "NPP"), col = c("green", "blue"), lty = 1)
```



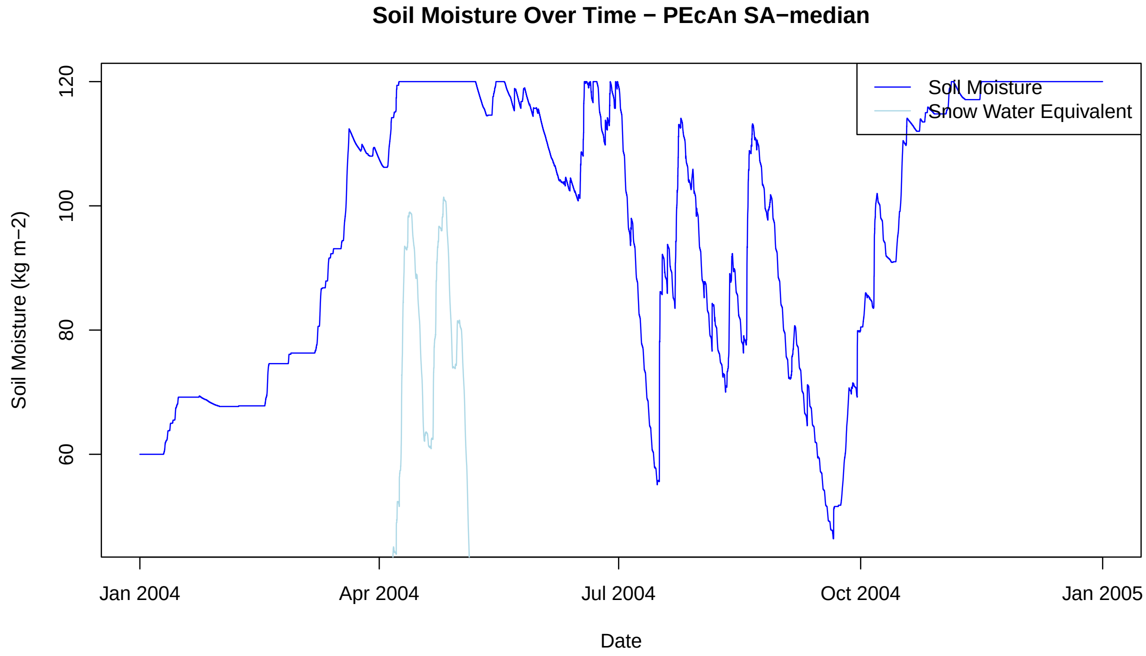
16.2 Plot Carbon Pools

```r
# Plot Total Live Biomass and Total Soil Carbon
plot(model_output$posix, model_output$TotLivBiom,
  type = "l",
  col = "darkgreen",
  xlab = "Date",
  ylab = "Carbon Pool (kg C m-2)",
  main = paste("Carbon Pools Over Time - PEcAn", runid)
)
lines(model_output$posix, model_output$TotSoilCarb, col = "brown")
legend("topright", legend = c("Total Live Biomass", "Total Soil Carbon"), col = c("darkgreen'
```



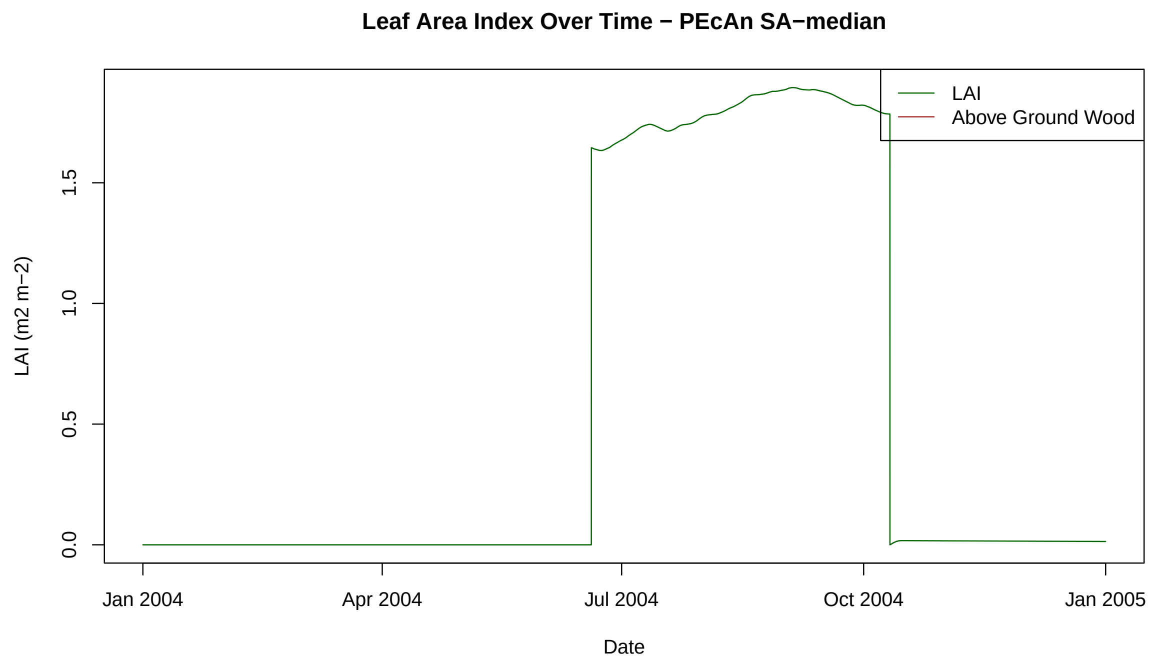
16.3 Plot Water Variables

```
# Plot Soil Moisture and Snow Water Equivalent
plot(model_output$posix, model_output$SoilMoist,
  type = "l",
  col = "blue",
  xlab = "Date",
  ylab = "Soil Moisture (kg m-2)",
  main = paste("Soil Moisture Over Time - PEcAn", runid)
)
lines(model_output$posix, model_output$SWE, col = "lightblue")
legend("topright", legend = c("Soil Moisture", "Snow Water Equivalent"), col = c("blue", "lig
```



16.4 Plot LAI and Biomass

```
# Plot Leaf Area Index (LAI) and Above Ground Wood
plot(model_output$posix, model_output$LAI,
  type = "l",
  col = "darkgreen",
  xlab = "Date",
  ylab = "LAI (m2 m-2)",
  main = paste("Leaf Area Index Over Time - PEcAn", runid)
)
lines(model_output$posix, model_output$AbvGrndWood, col = "brown")
legend("topright", legend = c("LAI", "Above Ground Wood"), col = c("darkgreen", "brown"), lty
```



14. Conclusion

This notebook demonstrated how to set up, run, and analyze a PEcAn ecosystem model workflow programmatically. You can now modify parameters, try different models, or extend the analysis as needed.

Try editing the `pecan.xml` file. Give it a new name and update the **settings_path** variable at the beginning of this Demo to point to the new file. See how the changes affect the model output!

15. Clean Up Workflow Output (Optional)

If you want to remove all files and directories created by this workflow and start fresh, you can run the following code. This will delete the entire output directory specified in your settings. **Use with caution!**

```
# WARNING: This will permanently delete all workflow output files!
# Uncomment the line below to enable cleanup.
# fs::dir_delete(settings$outdir)
```

17. Session Information

Print R session information for reproducibility.

```r
sessionInfo()
```

```
R version 4.5.1 (2025-06-13)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.2 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0  LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
 [9] LC_ADDRESS=C               LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Asia/Kolkata
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] PEcAn.SIPNET_1.9.1          PEcAn.all_1.9.0
 [3] PEcAn.workflow_1.9.0        PEcAn.remote_1.9.0
 [5] PEcAn.benchmark_1.7.4       PEcAn.priors_1.7.4
 [7] PEcAn.emulator_1.8.1        PEcAn.assim.batch_1.9.0
 [9] PEcAn.data.remote_1.9.0     PEcAn.data.land_1.8.1
[11] PEcAn.data.atmosphere_1.9.0 PEcAn.uncertainty_1.8.1
[13] PEcAn.utils_1.8.1           PEcAn.logger_1.8.3
[15] PEcAn.MA_1.7.4              PEcAn.settings_1.9.1
[17] PEcAn.DB_1.8.1

loaded via a namespace (and not attached):
 [1] gtable_0.3.6             xfun_0.52
 [3] ggplot2_3.5.2            lattice_0.22-5
```

```
 [5] vctrs_0.6.5                   tools_4.5.1
 [7] generics_0.1.4                tibble_3.3.0
 [9] proxy_0.4-27                  pkgconfig_2.0.3
[11] KernSmooth_2.23-26            RColorBrewer_1.1-3
[13] lifecycle_1.0.4               PEcAn.ModelName_1.8.1
[15] compiler_4.5.1                farver_2.1.2
[17] codetools_0.2-20              ncdf4_1.24
[19] htmltools_0.5.8.1             class_7.3-23
[21] yaml_2.3.10                   pillar_1.11.0
[23] PEcAn.visualization_1.8.1 classInt_0.4-11
[25] sessioninfo_1.2.3             iterators_1.0.14
[27] abind_1.4-8                   foreach_1.5.2
[29] tidyselect_1.2.1              digest_0.6.37
[31] stringi_1.8.7                 sf_1.0-21
[33] dplyr_1.1.4                   purrr_1.1.0
[35] labeling_0.4.3                rprojroot_2.1.0
[37] fastmap_1.2.0                 grid_4.5.1
[39] here_1.0.1                    cli_3.6.5
[41] magrittr_2.0.3                XML_3.99-0.18
[43] e1071_1.7-16                  withr_3.0.2
[45] scales_1.4.0                  timechange_0.3.0
[47] lubridate_1.9.4               rmarkdown_2.29
[49] gridExtra_2.3                 coda_0.19-4.1
[51] evaluate_1.0.4                knitr_1.50
[53] RPostgreSQL_0.7-8             rlang_1.1.6
[55] Rcpp_1.1.0                    glue_1.8.0
[57] DBI_1.2.3                     rstudioapi_0.17.1
[59] jsonlite_2.0.0                R6_2.6.1
[61] units_0.8-7
```